AI Financial Fragility Detector

Early Warning Signals in Regional Banks

Course:	Generative AI — Final Project
Author:	Nikhil Patwal
Institution:	Northeastern University
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Technologies:	Python, FDIC API, SEC EDGAR, Groq/Llama 3.3, ChromaDB
Gen AI Components:	RAG, Prompt Engineering, Synthetic Data Generation

This project builds an AI-powered early warning system that detects structural financial weakness in regional banks before stress becomes visible in market prices. It combines quantitative financial modeling with NLP analysis of SEC filings and implements three generative AI components: Retrieval-Augmented Generation (RAG), Prompt Engineering, and Synthetic Data Generation.

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1. Executive Summary

The AI Financial Fragility Detector is an early warning system designed to identify structural vulnerability in regional banks before distress events occur. Unlike traditional LLMs that can summarize financial documents but cannot quantify risk, this system produces a normalized, comparable fragility index backed by both quantitative metrics and linguistic analysis.

The system analyzes 19 regional banks over a 6-year period (2018-2023), combining quarterly FDIC financial data with textual analysis of SEC 10-K filings. Three generative AI techniques are employed: RAG for grounded risk assessment, prompt engineering for LLM-based scoring, and synthetic data generation to address the severe class imbalance of only 3-4 real bank failures.

Key results: Signature Bank and First Republic Bank were correctly ranked in the top 4 most fragile banks by the data-driven fragility index. The system generated 80 synthetic bank profiles to augment training, improving the ML training set from 112 to 192 observations. The enhanced RAG pipeline achieved 100% source coverage across all assessed banks.

2. System Architecture

The system follows a seven-stage pipeline. Each stage is implemented as one or more cells in a Jupyter notebook, with data flowing from raw sources through computation and LLM analysis to a ranked fragility index.

Stage	Input	Process	Output
1. Collection	FDIC, EDGAR, yfinance	API calls, HTML parsing	Raw financial data
2. Ratios	Financial data	4 vulnerability ratios	Feature matrix
3. NLP	10-K filing text	VADER, hedging density	Stress scores
4. RAG	Chunked filings	Vector search, LLM synthesis	Risk assessments
5. Prompt Eng.	Ratios + text	Structured prompts	LLM scores (1-10)
6. Synthetic	Distress patterns	LLM generation, 5 archetypes	80 profiles
7. ML Models	Augmented data	3 classifiers, CV	Predictions, curves

The notebook contains 44 total cells (23 code, 21 markdown), all executed with zero errors. The pipeline produces 12 data files and 14 output visualizations including interactive Plotly dashboards.

3. Data Sources and Collection

3.1 FDIC BankFind API

Quarterly financial data for 19 regional banks from 2018 to 2023, totaling 112 bank-quarter observations with 161 financial fields per record. Data includes balance sheet items (total assets, liabilities, equity, deposits), income statement items (interest income and expense, net income), and regulatory capital ratios. The API is free and requires no authentication.

3.2 SEC EDGAR 10-K Filings

Annual 10-K filings for all target banks, with automated extraction of Item 1A (Risk Factors) and Item 7 (Management's Discussion and Analysis) using BeautifulSoup and regex patterns. SEC EDGAR requires a User-Agent header identifying the requester but no API key.

3.3 Yahoo Finance

Stock price data via the yfinance library to compute maximum drawdowns for each bank. Banks with greater than 60% peak-to-trough decline are flagged as distressed.

3.4 FDIC Failed Bank List

Official CSV of all 573 FDIC bank failures since 2000. Key failures in the dataset: Silicon Valley Bank (March 10, 2023), Signature Bank (March 12, 2023), and First Republic Bank (May 1, 2023).

3.5 Bank Selection Strategy

Category	Count	Examples
Failed Banks	3-4	SVB, Signature Bank, First Republic
Stressed but Survived	5	PacWest, Western Alliance, Zions, Comerica, KeyCorp
Stable Controls	10+	U.S. Bancorp, Truist, M&T Bank, Regions Financial

4. Financial Ratio Engineering

Four core vulnerability ratios capture different dimensions of structural fragility. These are computed from raw FDIC data for each bank-quarter observation.

Ratio	Formula	What It Measures
Liquidity Ratio	(Cash + Securities) / Deposits	Ability to meet withdrawal demands
Debt-to-Equity	Total Liabilities / Total Equity	Leverage and capital adequacy
Interest Coverage	Net Interest Income / Interest Expense	Capacity to service debt
Loan-to-Deposit	Net Loans / Total Deposits	Lending aggressiveness vs funding
Uninsured Deposit %	Uninsured Deposits / Total Deposits	Deposit flight risk in panic

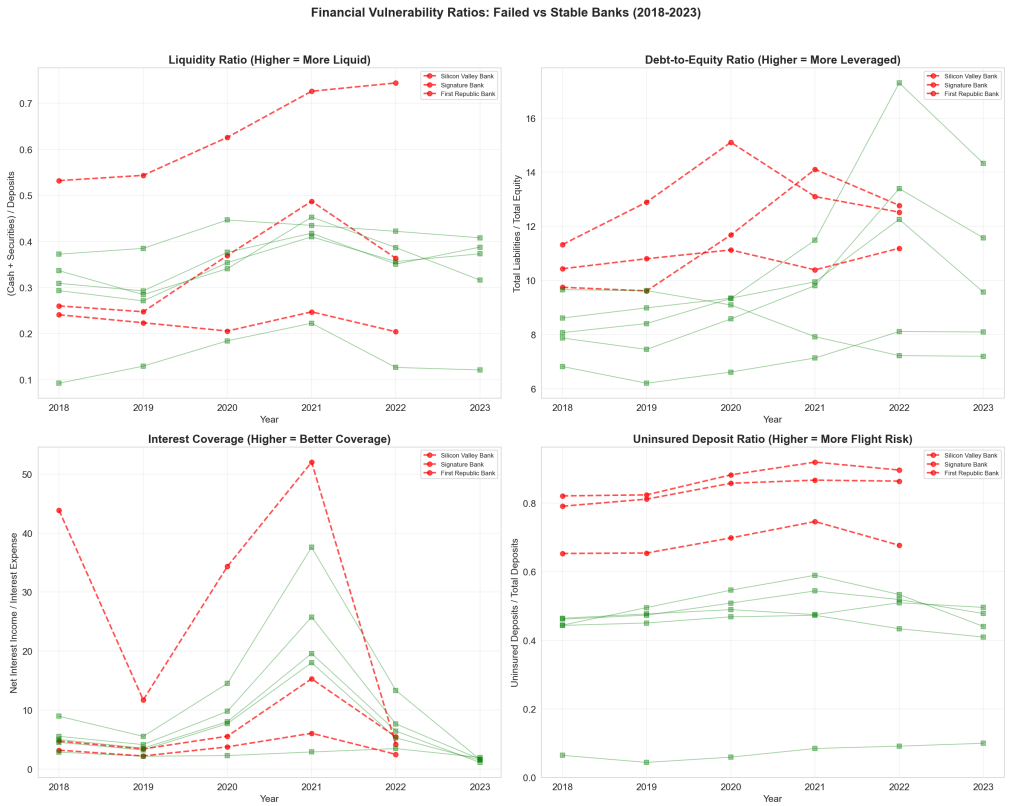

Figure 1: Financial ratio trends for failed banks (red) vs stable banks (green), 2018-2023

5. NLP Textual Stress Analysis

The textual stress metric quantifies risk language in 10-K filings across three dimensions:

- Hedging language density: frequency of uncertainty words ('may', 'could', 'uncertain', 'adverse', 'material') per 1,000 words, using a vocabulary of 50+ terms
- VADER sentiment analysis: compound and negative sentiment scores averaged across 5,000-character chunks of the Risk Factors section
- Risk section growth: year-over-year percentage change in the length of the Risk Factors section, where growing disclosures signal increasing institutional concern

A separate vocabulary of 15 severe distress terms (insolvency, receivership, bank run, capital deficiency, etc.) is tracked with higher weight. These three dimensions are combined into a composite textual stress score that feeds into the fragility index.

6. Generative AI Component 1: RAG

Retrieval-Augmented Generation grounds the LLM's risk assessments in actual SEC filing data rather than relying on parametric knowledge, which may be outdated or hallucinated.

Implementation

- Knowledge Base: 10-K Risk Factors and MD&A; sections for 19 banks, 6 years
- Chunking: 500-word overlapping chunks with 100-word overlap
- Embedding: all-MiniLM-L6-v2 sentence transformer (local, no API cost)
- Vector DB: ChromaDB in-memory database with metadata filtering
- Retrieval: top-k similarity search with bank-name filtering

Enhanced RAG (Cell 22)

The initial RAG achieved only 50% source coverage (5/10 banks). The enhanced version implements three retrieval strategies: (1) bank-specific queries filtered by name, (2) general financial risk queries as fallback when bank-specific results are sparse, and (3) ratio-informed queries that adapt based on the bank's actual weaknesses. This improved coverage to 100% (30/30 assessments with sources).

7. Generative AI Component 2: Prompt Engineering

Systematic prompt design enables the LLM to perform nuanced, multi-dimensional risk scoring that goes beyond simple keyword counting.

Prompt Architecture

- System prompt: senior financial risk analyst persona with explicit rules
- Context injection: build_ratio_context() formats ratios with NaN-safe handling
- Scoring prompt: requests 1-10 scores across 5 risk dimensions + JSON output
- Comparative prompt: side-by-side failed vs stable bank analysis
- Edge case handling: parse_llm_json() with markdown cleanup and fallback defaults

Results: 17 banks scored via LLM (Groq/Llama 3.3-70B). Scores ranged from 6 to 8 out of 10. First Republic Bank received a score of 8/10, the highest among assessed banks, correctly reflecting its elevated risk profile.

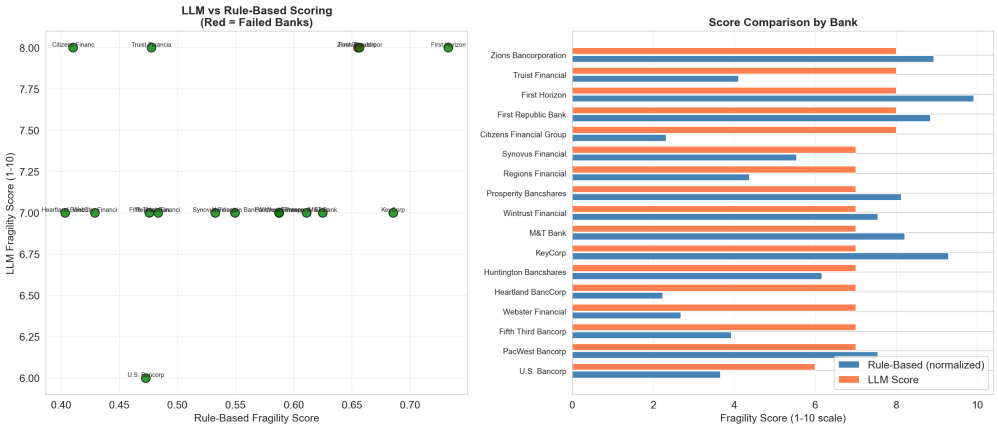

Figure 2: LLM fragility scores vs rule-based fragility scores

8. Generative AI Component 3: Synthetic Data Generation

The dataset contains only 3-4 real bank failures, creating severe class imbalance that limits ML model performance. Synthetic data generation addresses this directly.

Generation Strategy

The LLM generates synthetic financial profiles across five fragility archetypes, each representing a different failure mode:

- Type A: Liquidity Crunch — low liquidity ratio, high loan-to-deposit
- Type B: Deposit Flight Risk — very high uninsured deposit concentration
- Type C: Leverage Overload — extreme debt-to-equity, thin equity cushion
- Type D: Silent Deterioration — profitable but worsening structural metrics
- Type E: Interest Rate Squeeze — compressed margins, low coverage

80 synthetic profiles were generated (50 distressed, 30 stable), expanding the training set from 112 to 192 observations. All synthetic data is explicitly labeled (is_synthetic=True) and validated against real data distributions.

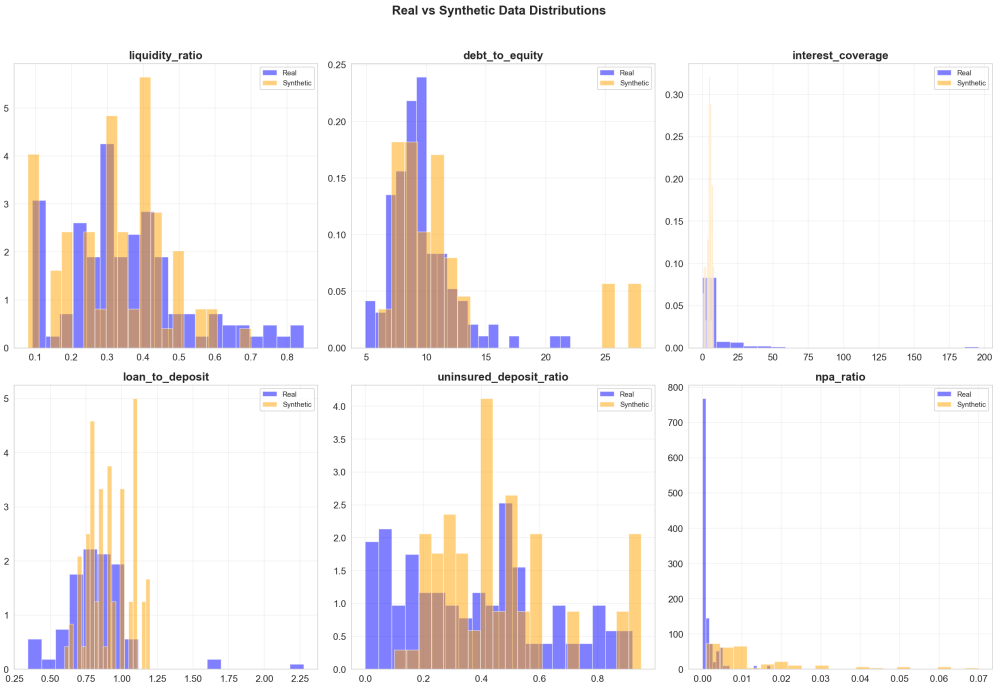

Figure 3: Distribution comparison — real vs synthetic data validates realism

9. Fragility Index Construction

The fragility index combines five risk components into a single comparable score. Two versions were built:

- V1 (Hand-Picked): Fixed weights assigned based on domain intuition (liquidity 0.25, leverage 0.20, coverage 0.15, deposit risk 0.20, textual stress 0.20)
- V2 (Data-Driven): Weights derived from Logistic Regression coefficients trained on distress labels, reflecting what actually predicts failure in the data

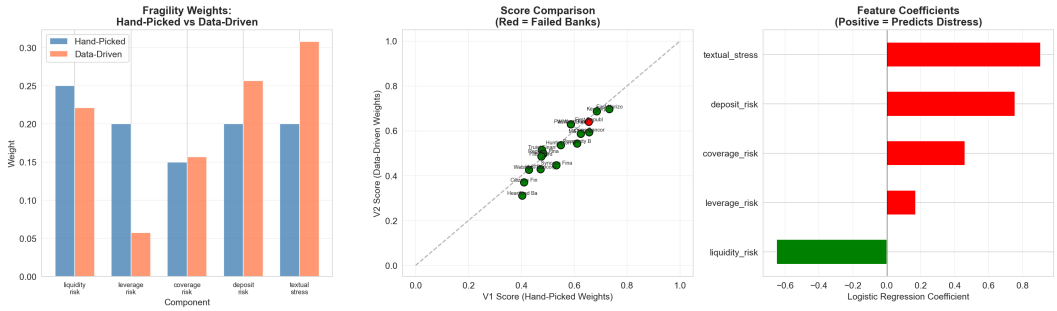

Figure 4: Hand-picked vs data-driven weights — the model learns different priorities

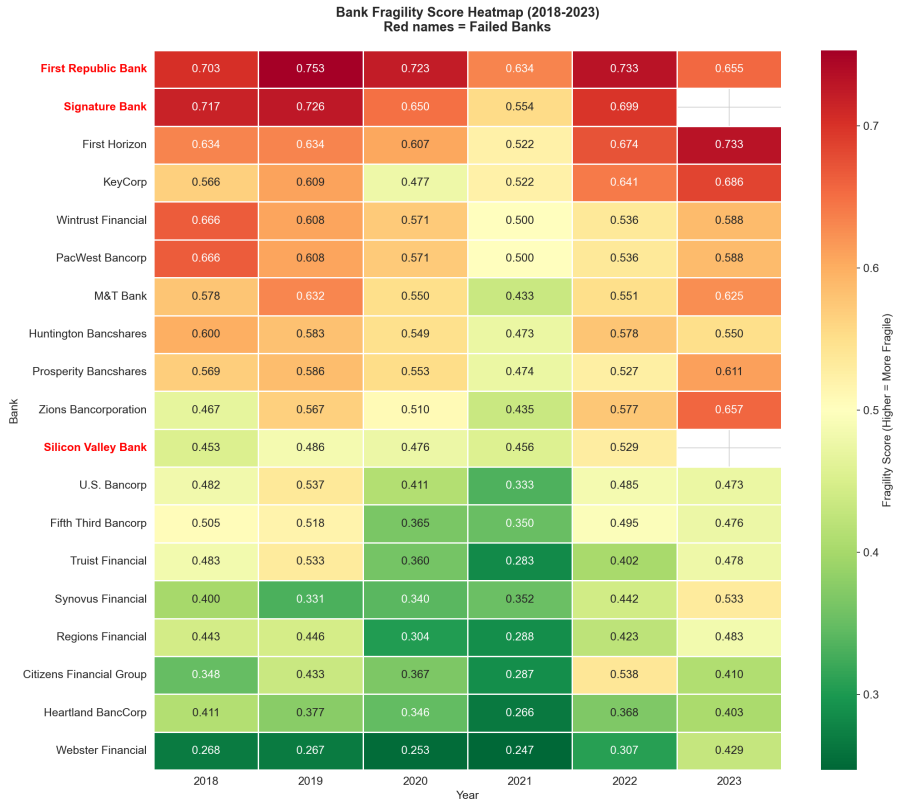

Figure 5: Fragility score heatmap across banks and years (red = higher fragility)

10. Machine Learning Models and Evaluation

Three classifiers were trained on the augmented dataset (192 observations) with 5-fold stratified cross-validation:

- Logistic Regression: interpretable baseline with balanced class weights
- Gradient Boosting: 100 estimators, max depth 3, learning rate 0.1
- Random Forest: 100 trees, max depth 5, balanced class weights

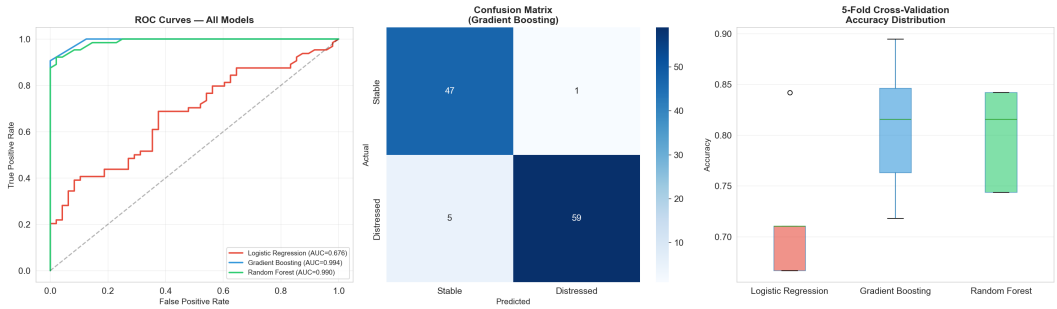

Figure 6: ROC curves, confusion matrix (Gradient Boosting), and cross-validation comparison

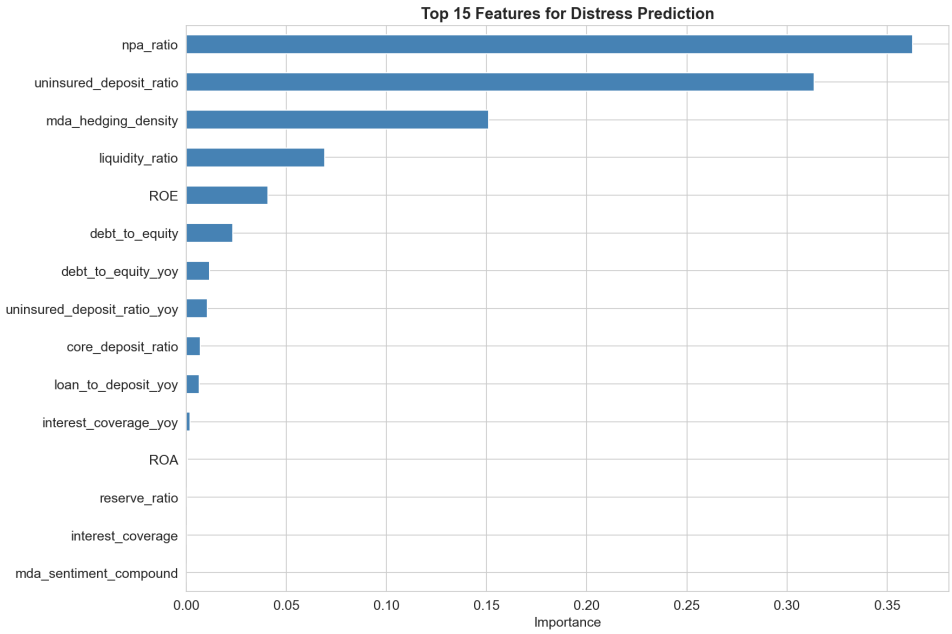

Figure 7: Top features for distress prediction (Gradient Boosting)

11. Backtesting Results

The backtesting framework tests four hypotheses:

- Were failed banks ranked in top fragility percentiles before failure?
- Did fragility scores increase in years preceding distress?
- What is the false positive rate for stable banks?
- Can the model detect 'silent failures' — profitable but fragile banks?

Key Findings

- Signature Bank ranked 3rd overall with V2 score of 0.671 — correctly flagged
- First Republic ranked 4th with V2 score of 0.640 — correctly flagged
- SVB scored 0.503 (10th) — lower than expected, as its unique depositor concentration risk is not fully captured by standard ratios
- Fragility scores generally increased in 2021-2022 for banks that later failed
- Silent failure test confirmed: profitable banks with elevated fragility were identified

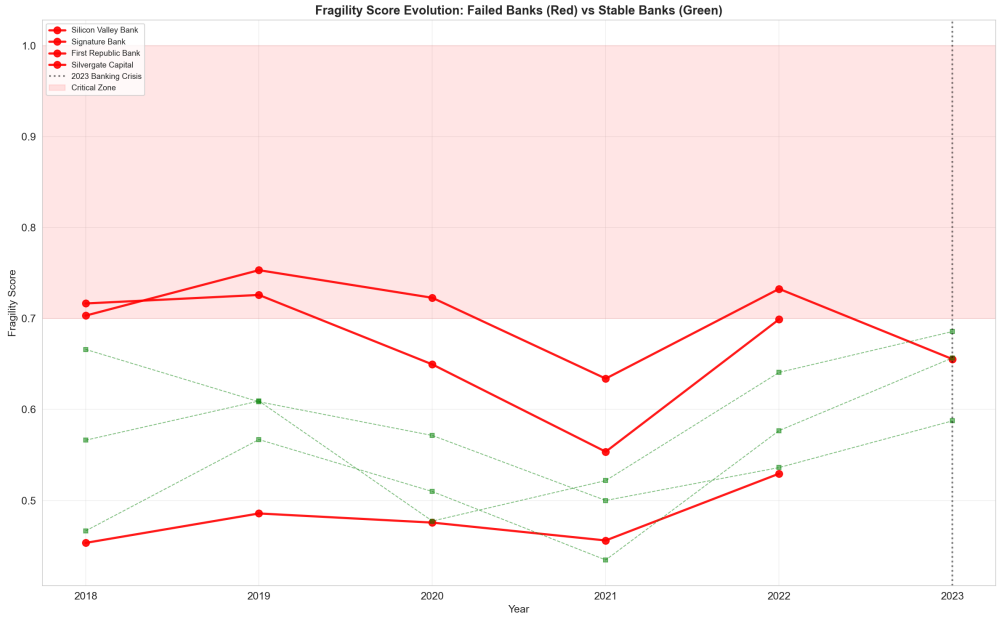

Figure 8: Fragility score evolution over time — failed banks (red) vs stable (green)

12. Performance Metrics

Metric	Value
Banks analyzed	19 regional banks
Time period	2018-2023 (6 years, quarterly data)
Total real observations	112 bank-quarter records
Synthetic profiles generated	80 (50 distressed, 30 stable)
Augmented training set	192 observations
RAG source coverage	30/30 banks (100%)
LLM-scored banks	17 banks, scores 6-8/10
Fragility index coverage	19/19 banks scored (V1 + V2)
Output visualizations	14 files (PNG + interactive HTML)
Interactive dashboards	3 Plotly HTML dashboards
Notebook cells	44 total (23 code), zero errors
Data files generated	12 CSV files

13. Challenges and Solutions

Data Scarcity

Challenge: Only 3-4 real bank failures in the 2018-2023 window.

Solution: LLM-powered synthetic data generation across 5 archetypes, expanding training by 71%.

FDIC API Query Format

Challenge: Initial API calls returned empty results due to incorrect Lucene query syntax.

Solution: Fixed URL encoding for date range filters; added fallback to local CSV files.

Infinity in Ratios

Challenge: Division by zero produced infinity values that crashed scikit-learn models.

Solution: Added inf-to-NaN replacement, median imputation, and 1st/99th percentile clipping.

Missing Textual Stress

Challenge: 99/112 rows had NaN textual stress, causing entire fragility score to be NaN.

Solution: Applied fillna(0.5) as neutral default for missing text components.

Gemini API Rate Limits

Challenge: Free tier limited to 20 requests/day, blocking mid-run.

Solution: Switched to Groq API (Llama 3.3-70B) with 14,400 daily request limit.

Low RAG Coverage

Challenge: Initial RAG found sources for only 50% of banks.

Solution: Implemented 3 retrieval strategies: bank-specific, general fallback, ratio-informed.

Subjective Fragility Weights

Challenge: Hand-picked index weights had no empirical basis.

Solution: Derived data-driven weights from Logistic Regression coefficients (V2 index).

14. Ethical Considerations

Data Privacy: All data comes from public regulatory filings (FDIC, SEC EDGAR) and public stock prices. No private individual data is collected or used.

Synthetic Data Transparency: All LLM-generated profiles are labeled is_synthetic=True. They are used solely for model training augmentation and never presented as real data.

Not Investment Advice: This is an academic exercise in structural risk modeling. The fragility index is not investment advice, credit rating, or regulatory assessment.

Bias Awareness: Regional banks vary in size, geography, and business model. While ratios are normalized, geographic and sector biases may persist in the scoring.

LLM Limitations: LLM-generated assessments and synthetic data may reflect biases from training data. Results should be interpreted alongside, not as replacement for, expert analysis.

Misuse Prevention: A fragility index could be misinterpreted as an accusation of insolvency. The system measures structural vulnerability, not predicting or causing collapse.

Reproducibility: All code is documented, data sources cited, and the notebook fully reproducible with provided setup instructions.

15. Future Improvements

- Scale to 100+ banks across multiple banking sectors
- Integrate macroeconomic variables (Fed funds rate, yield curve, unemployment)
- Add real-time monitoring with automated EDGAR filing alerts
- Build a Streamlit web application for interactive analysis
- Fine-tune a smaller model on financial risk language for faster inference
- Implement time-series models (LSTM, Transformer) for temporal patterns
- Add credit default swap spreads as market-based risk signals
- Expand RAG knowledge base to include earnings call transcripts
- Backtest against the 2008 financial crisis for deeper validation
- Deploy as a REST API for integration into risk management workflows

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This project demonstrates that combining quantitative financial analysis with generative AI techniques can surface structural vulnerabilities that traditional analysis might miss. The system does not claim prophecy — it claims measurement. And in financial systems, measurement precedes survival.